PHC-X: Provenance-aware Pre-Generation Hallucination Risk Prediction and Intervention for Multi-Turn Conversation

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Abstract

We present PHC-X, a framework that predicts the risk that the next assistant turn will hallucinate before any token is generated, and then chooses an intervention: generate, retrieve evidence, clarify, or abstain. Unlike white-box pre-decoding probes that read a single prompt's hidden states, PHC-X scores the upcoming turn from conversation-level black-box signals: cross-turn consistency (contradiction, entity drift, temporality, referential ambiguity, false presuppositions), the provenance of claims stored in conversation memory, and the coverage of retrieved evidence. A calibrated risk model maps these 54 features to a probability, and a cost-sensitive policy with monotone thresholds converts it into an action. On PHC-Conversation, a family-split, leakage-checked benchmark of 1100 prompts in 238 families (test $n = 161$), PHC-X reaches AUROC 0.984 [0.966, 0.996], AUPRC 0.995, F1 0.961, and ECE 0.026, versus AUROC 0.717 for a transparent heuristic baseline, with multi-seed AUROC 0.980±0.003 (seeds 42, 123, 2024). Action-level safety gates PASS: generated turns carry hallucination rate 0.021 against a base rate of 0.633, while 0.961 of all positives are caught and coverage is 0.509. Ablations attribute most of the gain to history, provenance, and evidence features (prompt-only AUROC 0.846). The system ships end-to-end: CLI, REST API, browser extension, and dashboard.

1 Introduction

Hallucination detection has largely followed a post-hoc pattern: generate, then check—self-consistency sampling [Manakul et al.(2023)], semantic entropy [Kuhn et al.(2023)], or verification chains [Dhuliawala et al.(2024)]. A complementary line predicts hallucination before decoding starts, but so far only by probing the model's internal states for a single prompt: FactCheckmate [Alnuhait et al.(2025)], KEEN [Gottesman and Geva(2024)], HALP [Kogilathota et al.(2026)], and probe-based pre-decoding detectors [Miftakhova and Zaytsev(2026), Guo et al.(2026)]. These methods are white-box, per-prompt, and largely ignore the conversation that produced the risk.

Meanwhile, multi-turn deployments accumulate risk across turns: stale or contradicted history, unverified claims copied forward from earlier assistant messages, entity and temporal drift, and presuppositions smuggled into follow-up questions. Dialogue-level benchmarks measure this afterwards [Chen et al.(2024), Fan et al.(2026)]; guardrail work scores accumulated session risk [Stateful Guardrails(2026)] or gates memory writes [Zhang and Li(2026)], but no existing system predicts the risk of the next turn and intervenes on it. HalluGuard explicitly lists conversation-level "prognostic indicators" as future work [Zeng et al.(2026)].

PHC-X fills this gap with four contributions:

1. Conversation-aware pre-generation risk prediction. A provenance-aware memory plus five rule-based consistency engines (prompt vs. history contradiction, entity drift, temporal inconsistency, false presuppositions, referential ambiguity) and evidence-coverage scoring feed a 54-dim feature vector computed before generation, entirely black-box (no model internals required).

2. Calibrated, cost-sensitive intervention. Risk probabilities are calibrated on a validation split (ECE 0.026 vs. 0.104 uncalibrated) and mapped to five actions by a monotone threshold policy with a budget-aware blocking gate.

3. PHC-Conversation benchmark. 1100 prompts in 238 near-duplicate families (splits 768/171/161), 11 categories (ten risk categories plus short benign first-turn prompts), union gold+verified labels with post-hoc agreement 0.752, and automated leakage checks (`leakage_passed`=true).

4. End-to-end system. `phc` CLI, REST API with decision logging, MV3 browser extension with a blocking gate, and a dashboard; all offline, deterministic, and seeded.

2 Related Work

Post-hoc detection. SelfCheckGPT scores sampling consistency [Manakul et al.(2023)]; semantic entropy clusters sampled meanings [Kuhn et al.(2023)] and can be probed cheaply [Kossen et al.(2024)]. Verification-then-revision pipelines (CoVe [Dhuliawala et al.(2024)], RARR [Gao et al.(2023)]) all operate after generation.

Pre-generation detection. FactCheckmate trains a hidden-state classifier and steers activations [Alnuhait et al.(2025)]; KEEN probes entity representations [Gottesman and Geva(2024)]; HALP probes pre-generative VLM states [Kogilathota et al soft-target attention probing [Miftakhova and Zaytsev(2026)] and HalluTracer [Guo et al.(2026)] predict truthfulness before decoding; confidence-aware routing estimates $P(\text{hallucinate} \mid q)$ before answering [Nandakishor(2025)]; HaluNet fuses internal signals [Tong et al.(2025)]; kernel language entropy scores semantic uncertainty over internal representations [Nikitin et al.(2024)]; simple factuality probes extend single-pass probing to long-form generation [Han et al.(2025)]; attention-sink dynamics offer a training-free internal signal [Binkowski et al.(2026)]. All are single-prompt and white-box.

Conversation-level risk. DiaHalu evaluates dialogue-level hallucination [Chen et al.(2024)]; HalluHard shows multi-turn error cascade [Fan et al.(2026)]; Stateful Guardrails accumulates session-level risk [Stateful Guardrails(2026)] ConsistencyGate admits memory writes via self-consistency [Zhang and Li(2026)]; STALE benchmarks stale-memory conflicts [Chao et al.(2026)]. False presuppositions in prompts are known to be adopted rather than rejected [Sieker et al.(2025)]. PHC-X is, to our knowledge, the first to predict the risk of the next turn from these conversation signals and act on it before decoding.

3 The PHC-X Framework

3.1 Provenance-aware conversation memory

Every user/assistant turn is appended to an immutable ConversationMemory. Claims extracted from assistant turns carry origin (`user`, `assistant`, `evidence`, `model`), a trust score, and a verification state. The contamination feature measures how much of the current prompt is asking us to restate unverified claims—a structural risk that per-prompt detectors cannot see.

3.2 Consistency engines

Five deterministic engines compare the pending prompt against memory and evidence: (i) contradiction (statement-vs-statement, skipping genuine questions), (ii) entity drift (typed slots replaced across turns, e.g., "migrated from X to Y"), (iii) temporal inconsistency (relative/absolute dates that conflict with stored facts), (iv) false presuppositions (unsupported change/existence/volume claims, with numeric normalization so that 70.0% matches 70.0), and (v) referential ambiguity (NP references resolving to multiple plausible antecedents; resolved references are not flagged). Each emits a bounded score in $[0,1]$.

3.3 Retrieval and evidence

A hashed n-gram TF index over the local evidence corpus retrieves the top passages; an evidence engine scores coverage, contradiction, staleness, and model-only support. Coverage above 0.3 multiplicatively dampens risk $(1 - 0.60 \cdot (\text{cov} - 0.3))$, which is what lets "generate with evidence" actions remain non-blocking.

Table 1: Test-split performance.

Method	AUROC	AUPRC	F1	Prec.	Rec.	ECE	Brier
PHC-X (RF, calibrated)	0.984	0.995	0.961	0.961	0.961	0.026	0.042
PHC-X (RF, uncalibrated)	0.988	0.993	0.923	0.857	1.000	0.104	0.058
Logistic regression (calibrated)	0.966	0.983	0.903	0.823	1.000	0.051	0.082
Heuristic (no training)	0.717	0.852	0.698	0.664	0.735	0.280	0.246

3.4 Features and risk model

`FeatureEngine` emits 54 features in six groups (prompt, history, provenance, consistency, evidence, model). Default serving is black-box: the model group is zeroed. We train random forests (default), logistic regression, and gradient boosting on the train split; probabilities are isotonic/Platt-calibrated on validation only, and thresholds are searched on validation only with a cost matrix that penalizes letting a positive turn through unblocked. The heuristic baseline—a weighted noisy-OR over explainable signals—is always available as a transparent fallback.

3.5 Intervention policy

Calibrated risk is mapped monotonically to actions at thresholds $\tau_{\text{low}} < \tau_{\text{mod}} < \tau_{\text{high}}$:

$$\text{action} = \begin{cases} \text{Generate} & r < \tau_{\text{low}} \\ \text{Generate_with_evidence} & \tau_{\text{low}} \le r < \tau_{\text{mod}} \\ \text{Retrieve} & \tau_{\text{mod}} \le r < \tau_{\text{high}} \\ \text{Clarify / Abstain} & r \ge \tau_{\text{high}} \end{cases}$$

with a user/latency budget deciding between clarify and abstain. Blocking actions (clarify/abstain) require an explicit user override in the extension; everything else proceeds, optionally after evidence injection.

4 PHC-Conversation Benchmark

Eleven categories (ten risk categories plus `short_benign`; the risk categories are `contradictory_context`, `entity_ambiguity`, `evidence_conflict`, `long_context_degradation`, `model_memory_contamination`, `referential_ambig` `supported_context`, `temporal_inconsistency`, `unsupported_context`, `unsupported_presupposition`) are generated from 238 families of near-duplicate prompt variants, so variants of one family never cross splits. Labels are the union of a template-derived gold label and a rule-verified label (source: `union(label_gold, label_verified)`); the two annotators agree on 0.752 of rows, which we report rather than hide. Leakage checks (duplicate prompts, semantic duplicates, family overlap, evidence overlap, variant crossing) all pass (`passed`=true). Label rates are 0.560/0.556/0.633for train/val/test.

5 Experimental Setup

All runs are seeded (seeds 42, 123, 2024); feature computation, calibration, and threshold search are separated by split to prevent leakage. Metrics: AUROC/AUPRC, F1 at the validation-tuned threshold (0.21), 10-bin ECE, Brier; 1,000-sample bootstrap CIs; DeLong (correlated) and exact McNemar tests; per-category breakdowns; action-level gates with coverage ≥ 0.500 and positives caught ≥ 0.500. Latency is measured locally on a warm pipeline: median 42.3 ms, p95 57.0 ms (including SHAP attribution).

6 Results

Headline. PHC-X attains AUROC 0.984 [0.966, 0.996], AUPRC 0.995 [0.981, 0.998], F1 0.961 (precision 0.961, recall 0.961), ECE 0.026, Brier 0.042, against AUROC 0.717 / AUPRC 0.852 / F1 0.698 for the

Table 2: Per-category test results.

Category	n	AUROC	F1	ECE
contradictory_context	12	0.950	0.947	0.083
entity_ambiguity	20	1.000	1.000	0.082
evidence_conflict	12	0.500	1.000	0.000
long_context_degradation	17	1.000	1.000	0.027
model_memory_contamination	12	1.000	1.000	0.057
referential_ambiguity	20	0.951	0.897	0.072
short_benign	11	0.500	0.000	0.000
supported_context	18	1.000	1.000	0.000
temporal_inconsistency	10	1.000	1.000	0.007
unsupported_context	18	0.785	0.833	0.192
unsupported_presupposition	11	1.000	1.000	0.000

Table 3: Feature ablations.

ID	Feature group	AUROC	Δ
F	Full black-box feature set	0.984	
G	Full + white-box model signals	0.981	-0.003
E	History + provenance + evidence	0.978	-0.006
D	Prompt + history + provenance + contradiction	0.895	-0.088
C	Prompt + history + provenance	0.889	-0.095
A	Prompt only	0.846	-0.138
B	Prompt + history	0.846	-0.138

heuristic baseline, and AUROC 0.966 for a logistic-regression probe on the same features. Calibration cuts ECE from 0.104 to 0.026 (Table 1). Multi-seed runs give 0.980±0.003.

Significance (honestly reported). The model and heuristic disagree on 61 test decisions (McNemar exact $p = <0.001$), yet their AUROCs are not significantly different under DeLong ($p = 0.460$) because the bootstrap CIs overlap on this test size. We report both rather than cherry-pick.

Ablations. Prompt-only features reach AUROC 0.846; adding history and consistency engines improve the score; the history+provenance+evidence group reaches the best single configuration (0.978); the full black-box model scores 0.984(Table 3). Conversation context—not the prompt alone—carries most of the signal.

Intervention behaviour. The tuned policy blocks 39.9% of turns (coverage 0.509) while catching 0.961 of all positive (hallucination-prone) turns; generated turns carry hallucination rate 0.021 versus base rate 0.633—a safety margin of 0.613. All gates report PASS (Tables 4 and ??). Errors concentrate in `unsupported_context` and `temporal_inconsistency`, the categories with the most lexically-diverse phrasings.

7 External Generalization and Validation Status

We evaluated the synthetic-trained checkpoint without retraining or recalibration on two public datasets. On all 1103 DiaHalu records, AUROC was 0.491; on 20,000 paired HaluEval dialogue responses, AUROC was 0.500. The HaluEval pair shares the same prompt, history, and evidence for its correct and hallucinated responses, so a strictly pre-generation system cannot distinguish the pair by construction; this is a label-alignment stress test, not evidence that the model should predict an already-generated answer. HalluHard seed questions were used for unlabeled coverage only. These results show that PHC-Conversation performance does not establish cross-dataset generalization. The local LLM pilot uses explicitly labeled lexical approximations of SelfCheckGPT and semantic entropy on 20 DiaHalu rows per model; the wide intervals are exploratory, not publishable baseline claims. CoVe and RARR are post-generation verification/revision methods and are therefore reported as non-comparable to the pre-generation classifier.

Table 4: Action-level safety gates.

Gate	Value	Status
Coverage	0.509	PASS
Positives caught	0.961	PASS
Safety margin	0.613	–
Blocked share	0.491	–

Table 5: External generalization without retraining.

Dataset	n	AUROC	95% CI	AUPRC
DiaHalu	1103	0.491	0.481–0.499	0.472
HaluEval dialogue	20000	0.500	0.494–0.507	0.500

8 Deployment

CLI. Eleven `phc` subcommands cover dataset generation, training, evaluation, calibration, benchmarking, ablations, significance testing, reporting, single-prompt scoring, and serving. API. FastAPI with `/v1/analyze`, `/v1/batch`, `/v1/policy`, `/v1/metrics`, append-only decision logging, and warm-up so steady-state latency stays at 42.3 ms. Extension. A Manifest V3 extension intercepts Enter/send on any chat page, computes risk against the local API, shows a non-blocking panel for generate/retrieve, and a hard gate for clarify/abstain (with an explicit "send anyway"); offline it fails open after a timeout. Dashboard. Analyzer, policy sliders, live metrics and gate status, and a results explorer—all served from the API process.

9 Limitations and Ethical Considerations

(1) PHC-Conversation is our synthetic benchmark with a scripted assistant; labels are audited (agreement 0.752) but it is not an external community benchmark. (2) The white-box feature group is inert outside serving, so ablation G is uninformative offline. (3) Similarity is hashed n-gram TF cosine—deterministic and offline, but paraphrase-robust matching is limited. (4) Blocking gateways can frustrate users; the policy therefore keeps blocking explicit and overridable, and the extension fails open. (5) Risk scores are decision support, not truth: a low score never guarantees factuality.

10 Conclusion

PHC-X moves hallucination risk prediction from "after the fact" (and from the model's internals) to before generation, from the conversation. Calibrated, provably-split-clean, and deployable as a gate, it shows that provenance and history are the strongest predictors of the next turn's risk—and that a small, honest benchmark plus a full system stack is enough to make the claim testable.

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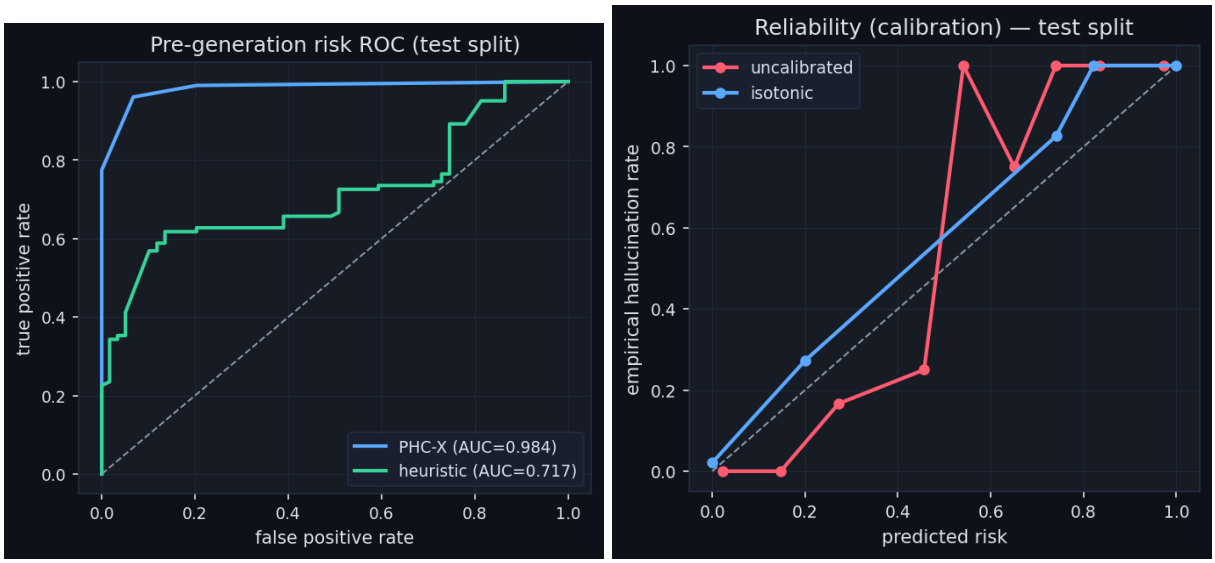


Figure 1: Left: ROC on the test split (PHC-X vs. heuristic baseline). Right: reliability before/after calibration.

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